

Today's Learning Record

Topic

TESTING, DEBUGGING, EXCEPTIONS, ASSERTIONS

Date

2026-09-22

① Check My Understanding

Questions completed: 5 / 5

Answers revised after AI hints: 0 / 5

② My Misconception

Before: I thought...

I did not have a major misconception going into the quiz, but I wanted to clarify the distinction between exceptions, exception handling, and assertions.

Now: I understand...

Testing is used to reveal whether a program has problems, while debugging investigates how an unexpected result happened and how to fix it. Path-complete glass-box testing does not guarantee that a program is bug-free because particular boundary cases can still fail. I also understand that exception handling refers to responding to an exception, rather than the exception or assertion itself. Assertions are mainly used as defensive programming to detect violated assumptions and halt execution.

③ Challenge the AI

One AI-generated question I challenged:

Question 2: "A test suite that is path-complete in glass-box testing guarantees that the program contains no bugs, because every possible path through the code has been tested."

Why?

☐ Ambiguous

☐ Oversimplified

☐ Technically questionable

☒ Too easy

☐ Other: _____

Brief explanation:

The question was too easy because the lecture directly gave the same concept and even provided an example showing that a path-complete test suite can still miss a bug. It tested recall of the lecture example more than conceptual reasoning. I also pointed out that the proposed replacement question was not ideal because it revealed the intended answer too directly.

④ One-Minute Reflection

One thing I am still unsure about:

When using `raise <exceptionName>(<arguments>)`, I want to understand the constraints on `<exceptionName>`: whether only built-in exception types can be raised or whether I can define my own custom exception. I learned that custom exceptions can be created by subclassing `Exception`, but this goes beyond what today's lecture explicitly covered.